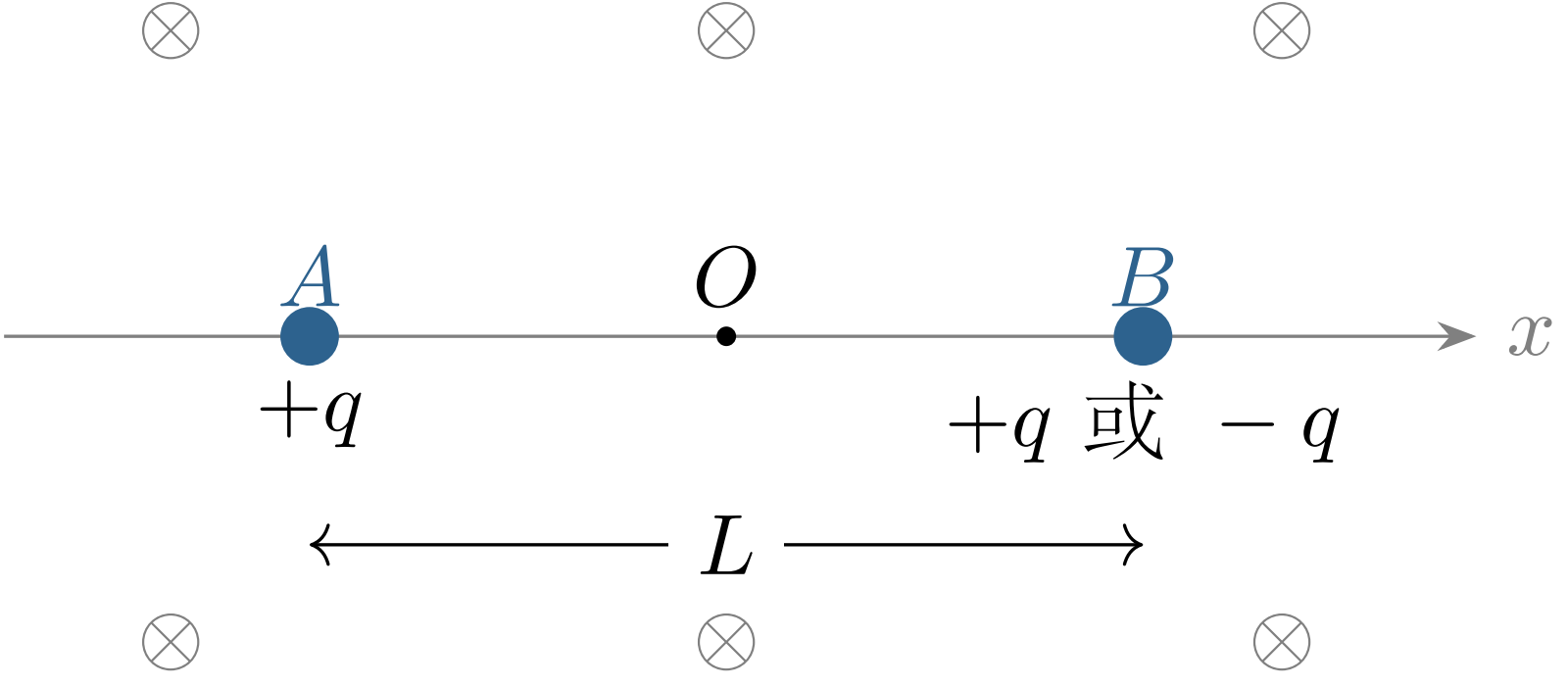$$\boldsymbol{B} = -B\boldsymbol{e}_z, \quad \boldsymbol{v}_A(0) = \boldsymbol{v}_B(0) = 0$$